Corrigé — Exercice 14 • Bac contrôle 2025

1) $\det(A) = 2\begin{vmatrix} 0 & 1 \\ 10 & -111 \end{vmatrix} - 1\begin{vmatrix} -1 & 1 \\ 101 & -111 \end{vmatrix} + 1\begin{vmatrix} -1 & 0 \\ 101 & 10 \end{vmatrix} = 2(-10) - (10) + (-10) = -40 \neq 0 : A$ inversible.

$A \times B = 40\, I_3$, donc $A^{-1} = \dfrac{1}{40}B$.

2) (S) s'écrit $A\,X = \begin{pmatrix} 9 \\ 3 \\ -353 \end{pmatrix}$, d'où $X = \dfrac{1}{40}B\begin{pmatrix} 9 \\ 3 \\ -353 \end{pmatrix} = \dfrac{1}{40}\begin{pmatrix} 80 \\ 0 \\ 200 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$.

$$(x, y, z) = (2,\ 0,\ 5)$$

3) $(a, b, c) = (2, 0, 5)$, donc $N = \overline{abac} = \overline{2\,0\,2\,5} = \boxed{2025}$.